Joint BBN and X-ray Constraints on keV Sterile-Neutrino Dark Matter, 2024–2026

HEP-Genesis-Agent session record
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Abstract

A three-wave interactive session surveyed 2024–2026 arXiv work on keV sterile-neutrino dark matter and combined tool-computed big-bang nucleosynthesis observables (Y_p, D/H, $\Delta N_{\rm eff}$) with published X-ray radiative-decay limits to map viable $(m_s, \sin^2 2\theta)$ regions. The sterile-neutrino BBN/X-ray bridge server was exercised with single-point predictions, product grids and 80×80 to 100×100 scans in $m_s = 1$ keV–100 keV and $\sin^2 2\theta = 10^{-15}$–$10^{-6}$, plus X-ray line predictions $E_\gamma = m_s/2$, Γ and fluxes for five targets, and literature figures were extracted. BBN alone does not constrain keV dark matter at X-ray-allowed mixings: along the Dodelson-Widrow dark-matter line $\Delta N_{\rm eff} = 10^{-4}$–$1.1 \times 10^{-2}$ and $|\Delta\chi^2_{\rm BBN}| < 0.15$, and the overclosure bound is 40–5000 times stronger than BBN. The Dodelson-Widrow mechanism cannot supply all dark matter; it is excluded by X-ray limits and independently by structure formation. Viable regions quoted in recent literature require resonant Shi-Fuller production with lepton asymmetry $L \gtrsim 0.5$ at $T > 20$ MeV at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$, flavor-dependent asymmetries, or non-oscillation reheating production with branching ratio $\lesssim 10^{-4}$. The server cannot test the large-L claim: its Shi-Fuller mode caps L at 0.1 and uses a heuristic $\Delta N_{\rm eff} = 0.1\times$ Dodelson-Widrow value. Tool budget exhaustion left the 7.1/10/20/40/50 keV X-ray limit table, the Dodelson-Widrow normalization settlement and BBN robustness reruns incomplete, so the joint exclusion remains anchored to prior scans and server heuristics with explicit caveats.

1 Introduction

The session addressed one scientific question, asked in the same form in two successive turns and then deepened in a third. The user requested a survey of recent arXiv work from 2024–2026 on sterile-neutrino dark matter, combined big-bang nucleosynthesis constraints using the sterile-neutrino tools for Y_p and D/H numbers with X-ray constraints, and a quantified statement of which $(m_s, \sin^2 2\theta)$ regions remain viable, with tool calls where possible. The third turn instructed to read the research board, identify the two weakest claims by missing evidence or thinnest tool support, deepen each by rerunning the relevant emulator at a contrasting parameter point and fetching one additional paper per weak claim, posting corrected findings, and writing an updated report. The scope therefore evolved from an initial broad survey and BBN plus relic-density mapping, through X-ray decay-line quantification and production-mechanism literature

review, to targeted deepening of the X-ray limit table and the production-mechanism and structure-formation mapping, with provenance and robustness checks.

2 Methods

All quantitative BBN and X-ray numbers were produced with the Sterile Neutrino → BBN/X-ray Bridge Server (serve-sbbn.sh) under local execution. The server maps ($m_s, \sin^2 2\theta$, mechanism) to $\Delta N_{\rm eff}$, $N_{\rm eff}$, $\Omega_s h^2$, Y_p, D/H, Li/H, $\chi^2_{\rm BBN}$, $\Delta\chi^2_{\rm BBN}$, lifetime τ, free-streaming scale and BBN-exclusion and overclosure flags, and to radiative-decay line energy $E_\gamma = m_s/2$, rate Γ and flux for Perseus, M31, Milky Way center, Coma and stacked galaxies. BBN abundances come from a fit to the PRyMordial network ($< 0.3\%$ error) whose only BBN input is $\Delta N_{\rm eff}$. The fit is calibrated for $N_{\rm eff} \in [2.044, 6.044]$, $\omega_b h^2 \in [0.019, 0.0257]$ and $\tau_n \in [866.4, 890.4]$ s and is valid for $m_s \lesssim 100$ keV where the sterile is relativistic at BBN; for $m_s \gtrsim 1$ MeV decay-product injection is not modeled and results are flagged not as constraints. Exclusion is $\Delta\chi^2 > 3.84$ (95% for one degree of freedom) versus the Standard Model. Observational baselines hard-coded in the server are $Y_p = 0.2449 \pm 0.004$ from Aver, Olive and Skillman JCAP 07 (2021) 029, D/H= $(2.547 \pm 0.025) \times 10^{-5}$ from Cooke, Pettini and Steidel ApJ 855 (2018) 102, $N_{\rm eff,CMB} = 2.99 \pm 0.17$ from Planck 2018 TT,TE,EE+lowE+lensing+BAO, Standard Model $N_{\rm eff} = 3.044$ and CMB-S4 forecast 3.044 ± 0.03. X-ray limits in the server are ten tabulated NuSTAR and XMM-Newton limits across the five targets, evaluated as whole-halo $f_{\rm DM} = 1$ with no field-of-view cut. Mechanisms are Dodelson-Widrow non-resonant and Shi-Fuller resonant, where the Shi-Fuller $\Delta N_{\rm eff}$ is a heuristic $0.1\times$ the Dodelson-Widrow value and the relic density is a fitting formula in L, capped at $L = 0.1$ with no ξ_e modeling; non-thermal reheating and inflaton-decay production with branching ratio $\sim 10^{-4}$, reheating temperature and inflaton mass are not modeled.

Tools exercised were `predict_sterile_nu`, `predict_sterile_nu_grid`, `predict_sterile_nu_xray_line`, `scan_sterile_nu_parameters`, `check_sterile_nu_sbl_anomaly`, `plot_sterile_nu_bbn_impact`, `plot_sterile_nu_exclusion`, `describe_sterile_nu_tools` and `list_sterile_nu_skills` on serve-sbbn.sh; `list_agent_files`, `list_files`, `list_input_files`, `list_datasets`, `get_dispatch`, `search_arxiv`, `read_text_file` and `get_paper_figures` on hep-research-aid; and `search_arxiv_abstracts`, `get_arxiv_metadata`, `download_arxiv_full_text`, `read_arxiv_chunk`, `web_search` and `cite` locally. Literature was located with `search_arxiv_abstracts`, `search_arxiv` and `web_search`, and full texts were downloaded for arXiv:2507.18752, arXiv:2405.17861, arXiv:2510.24560, arXiv:1702.08430, arXiv:1705.01837 and astro-ph/0501562. Figures were extracted with `get_paper_figures` from arXiv:2405.17861, arXiv:1705.01837, astro-ph/0501562 and astro-ph/0602028.

Three run directories were populated. The first, `runs/261021_110150`, contains a 100×100 Dodelson-Widrow scan `scan_DW_dc7743.csv/.npz`, BBN versus $N_{\rm eff}$ plot `bbn_vs_neff.png`, exclusion-plane plot `exclusion_scan_DW_dc7743.png`, six sterile-neutrino grids for Dodelson-Widrow and Shi-Fuller, `references.bib` and six paper PDF/TXT pairs. The second, `runs/261102_163543`, contains 6400-point and 400-point Dodelson-Widrow scans, 70-point and 400-point product grids, a 36-point reproducible Y_p/D/H grid, BBN versus $N_{\rm eff}$ plot and PDF, and two figures from arXiv:1705.01837. The third, `runs/261102_164917`, contains three grids at 40–50 keV and 20–50 keV for Dodelson-Widrow and Shi-Fuller and four figure directories. The supplement generated with this report holds the full tool-call log, board state, extra figures and code index.

3 Results

3.1 First wave: literature survey and BBN plus relic-density mapping

Four arXiv identifiers were resolved by metadata lookup to real papers: arXiv:2507.18752 (Vogel, Garcia Escudero, Froustey and Abazajian 2025, Shi-Fuller revival), arXiv:2502.17374 (Gorbunov, Kalashnikov and Krugan 2025, flavor-dependent asymmetry), arXiv:2601.03346 (Cline and Xu 2026, reheating and inflaton-decay production with branching ratio $\lesssim 10^{-4}$) and arXiv:2609.31340 (Gorbunov and Kalashnikov 2026, saturation of resonant production at high flavor asymmetry). Only arXiv:2507.18752 was read in full text; the other three were seen as abstracts, so their claims are reported as abstract-level. A contested finding from the full text states that lepton asymmetries $L \gtrsim 0.5$ at high temperature open significant viable parameter space at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$, compatible with X-ray constraints from NuSTAR at $\sin^2 2\theta \lesssim 10^{-12}$ in 6–40 keV and from INTEGRAL/SPI at lifetimes below $\sim 10^{20}$ s excluded at 95% corresponding to $\sin^2 2\theta \lesssim 3 \times 10^{-14}$ above ~ 40 keV, and with structure-formation bounds $m_{\rm th} > 8.4$ keV from galaxy counts at 95% and $m_{\rm th} > 8.5$ keV from counts plus strong lensing, while also quoting a Lyman-α preference $m_{\rm th} = 4.1^{+33.3}_{-1.5}$ keV and that the original Dodelson-Widrow mechanism is ruled out as producing all dark matter. A second contested finding maps mechanisms from literature quotes: Shi-Fuller with $L \gtrsim 0.5$ above 20 MeV viable at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$ with the NuSTAR and SPI ceilings above, flavor-dependent asymmetries with inverted ordering opening lower mixings and colder spectra that reopen lighter m_s, large L saturating resonant production, and reheating production opening new regions at very small mixing.

The server's radiative decay rate was verified against $\Gamma = 1.38\times10^{-32}\,{\rm s}^{-1}\,(\sin^2 2\theta/10^{-10})(m_s/{\rm keV})^5$ to within about 1.5%. At $m_s = 7.1$ keV and $\sin^2 2\theta = 10^{-10}$ the tool gives $\Gamma = 2.46 \times 10^{-28}$ s^{-1} versus 2.49×10^{-28} s^{-1} from the formula with $E_\gamma = 3.55$ keV; at 20 keV and 10^{-12} it gives 4.36×10^{-28} s^{-1} versus 4.42×10^{-28} s^{-1} with $E_\gamma = 10$ keV. The tool's built-in X-ray limits from NuSTAR M31 (Roach et al. 2020) and XMM-Newton blank sky (Dessert et al. 2020) exclude both points.

A Dodelson-Widrow benchmark grid at $m_s = 7.1, 10, 20, 50$ keV with $\sin^2 2\theta = 10^{-10}, 10^{-12}, 10^{-14}$ shows no BBN exclusion: $\Delta N_{\rm eff} \leq 2.9 \times 10^{-3}$ and $|\Delta\chi^2| \leq 0.04$. The relic density reaches $\Omega_s h^2 = 0.12$ only near $\sin^2 2\theta \sim 10^{-10}$ at 10–20 keV (0.087 at 10 keV, 0.30 at 20 keV which overcloses, 1.58 at 50 keV and 10^{-10}); at X-ray-allowed mixings 10^{-12}–10^{-14} the Dodelson-Widrow density is ≤ 0.016, so a non-Dodelson-Widrow mechanism is needed for all dark matter.

A 100×100 Dodelson-Widrow scan from 1–100 keV and $\sin^2 2\theta = 10^{-15}$–$10^{-6}$ finds BBN at $\Delta\chi^2 > 3.84$ excludes only $\sin^2 2\theta$ above 3.5×10^{-7} at 1 keV, 1.2×10^{-7} at 3.7 keV, 8.1×10^{-8} at 7 keV, 4.3×10^{-8} at 13.5 keV, 1.5×10^{-8} at 50 keV and 10^{-8} at 100 keV. The $\Omega_s h^2 = 0.12$ contour lies at $\sin^2 2\theta = 8.7 \times 10^{-9}$ at 1 keV, 2.5×10^{-9} at 2 keV, 1.2×10^{-9} at 3 keV, 4.8×10^{-10} at 5 keV, 2.55×10^{-10} at 7.1 keV, 1.37×10^{-10} at 10 keV, 3.9×10^{-11} at 20 keV, 7.6×10^{-12} at 50 keV and 2.2×10^{-12} at 100 keV, where $\Delta N_{\rm eff} = 10^{-4}$–$1.1 \times 10^{-2}$ and $|\Delta\chi^2_{\rm BBN}| < 0.15$. The overclosure bound is therefore 40 times stronger than BBN at 1 keV and about 5000 times stronger at 100 keV, and BBN leaves open all of the parameter space that Dodelson-Widrow dark matter actually occupies.

Tool-computed BBN at representative points, compared with the brief's $Y_p = 0.245\pm0.003$ and D/H$= (2.53 \pm 0.03) \times 10^{-5}$, gives Standard Model $N_{\rm eff} = 3.044$ with $Y_p = 0.246815$ ($+0.6\sigma$) and D/H$= 2.4433 \times 10^{-5}$ (-2.9σ). At 7.1 keV and 3×10^{-10}, $Y_p = 0.24684$, D/H$= 2.4440 \times 10^{-5}$ and $\Delta\chi^2 = -0.024$; at 1 keV and 3×10^{-7}, $\Delta N_{\rm eff} = 0.387$, $Y_p = 0.25193$ ($+2.3\sigma$), D/H$= 2.5718 \times 10^{-5}$ ($+1.4\sigma$) and $\Delta\chi^2 = +0.44$ allowed; at 10 keV and 10^{-7}, $\Delta N_{\rm eff} = 0.813$, $Y_p = 0.25730$ ($+4.1\sigma$), D/H$= 2.7145 \times 10^{-5}$ ($+6.2\sigma$) and $\Delta\chi^2 = 13.4$ excluded. The server's Dodelson-Widrow relic density scales as $\Omega_s h^2 = 0.1\,(\sin^2 2\theta/10^{-9})(m_s/3\,{\rm keV})^{1.8}$, checked at 3 keV and 10^{-9} giving 0.100

and at 10 keV and 10^{-9} giving 0.873 for ratio $8.73 = (10/3)^{1.8}$, but the normalization is three times higher than the brief's $0.1\,(\sin^2 2\theta/3\times10^{-9})(m_s/3\,\mathrm{keV})^{1.8}$, so the server's dark-matter line sits at three times smaller mixing.

The server cannot directly test the large-L Shi-Fuller claim. Its Shi-Fuller mode caps lepton asymmetry at 0.1 while the paper needs $L \gtrsim 0.5$, its $\Delta N_{\rm eff}$ is a heuristic 0.1× the Dodelson-Widrow value, and the asymmetry is not fed into the BBN network. At $m_s = 10, 20, 30$ keV with $\sin^2 2\theta = 10^{-14}$ it gives $\Delta N_{\rm eff} \sim 10^{-8}$ and $\Delta\chi^2_{\rm BBN} = 0$ allowed, with $Y_p = 0.24682$ and D/H$= 2.4433\times10^{-5}$ at 20 keV, while $\Omega_s h^2 = 0.122, 0.490, 1.10$ for $L = 10^{-3}$ and $1.22, 4.90, 11.0$ for $L = 0.1$, all flagged as overclosing.

A cross-check at the BEST short-baseline best fit $m_s = 1.14$ eV and $\sin^2 2\theta = 0.42$ gives $\Delta N_{\rm eff} = 1$ ($N_{\rm eff} = 4.044$), $Y_p = 0.25956$, D/H$= 2.777\times10^{-5}$ and $\Delta\chi^2_{\rm BBN} = 23.1$ excluded, 6.2σ tension with Planck $N_{\rm eff} = 2.99 \pm 0.17$ and 33σ at CMB-S4, with $\Omega_s h^2 = 29.3$ under Dodelson-Widrow.

Two contested X-ray findings were recorded. XRISM Resolve stacked 3.75 Ms on ten galaxy clusters (arXiv:2510.24560) and found no unidentified lines over 2.5–15 keV; its 3σ upper limit on the decay rate for $m_s \sim 7.1$ keV ($E = 3.55$ keV) is $\Gamma \sim 1.0\times10^{-27}$ s^{-1}, 3–4 times stronger than Hitomi Perseus but still five times above the Bulbul et al. 2014 XMM-Newton detection, so XRISM does not yet rule out the 3.55 keV line, scaling to $\sin^2 2\theta \lesssim 4\times10^{-10}$ at 7.1 keV from the server rate 2.46×10^{-28} s^{-1} at 10^{-10}. Krivonos, Barinov, Mukhin and Gorbunov (arXiv:2405.17861) use 11 years of NuSTAR stray-light data on the Milky Way to set 95% upper limits on $\sin^2 2\theta$ for $m_s \sim 6$–40 keV (photon energies 3–20 keV) plotted in their Fig. 3 over 10^{-15}–10^{-10}, stating the limits reach the bottom edge of realistic production models, with Eq. 2 $\Gamma_\gamma = 1.36\times10^{-22}(m_s/\mathrm{keV})^5 \sin^2 2\theta$ s^{-1} which equals 1.36×10^{-32} s^{-1} at $\sin^2 2\theta = 10^{-10}$, 1.4% below the 1.38×10^{-32} normalization and about 1% above the server.

A contested X-ray limit table at $7.1, 10, 20, 40, 50$ keV was assembled: at 7.1 keV XRISM gives $\Gamma < 1.0\times10^{-27}$ s^{-1} at 3σ converting to $\sin^2 2\theta \lesssim 4\times10^{-10}$ while NuSTAR gives $\lesssim 10^{-12}$ over 6–40 keV and the tool's built-in tables exclude 10^{-10}; at 10 keV NuSTAR $\lesssim 10^{-12}$ with tool $\Gamma = 1.36\times10^{-29}$ s^{-1}, $E = 5$ keV and $\tau = 7.35\times10^{28}$ s at 10^{-12} not excluded by the tool; at 20 keV NuSTAR $\lesssim 10^{-12}$ but the tool's tables exclude 10^{-12} so its NuSTAR M31 table is stronger than the summary value; at 40–50 keV INTEGRAL/SPI is quoted as $\lesssim 3\times10^{-14}$ with lifetime $> 10^{20}$ s. A contested final viability table states Dodelson-Widrow is not viable as all dark matter with the dark-matter line at $\sin^2 2\theta = 2.55\times10^{-10}$ at 7.1 keV, 1.37×10^{-10} at 10 keV, 3.9×10^{-11} at 20 keV and 7.6×10^{-12} at 50 keV (server normalization) all above XRISM, NuSTAR and SPI limits, binding constraint X-ray while BBN gives $\Delta\chi^2 < 0.15$ along the line; Shi-Fuller with $L \gtrsim 0.5$ is quoted viable at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$; flavor-dependent asymmetries and reheating production open further regions but rest on abstracts only.

The Dodelson-Widrow normalization dispute was quantified against the published fit in arXiv:1705.01837 Eq. 37, $m_s \sim 3.40\,\mathrm{keV}\,(\sin^2 2\theta/10^{-8})^{-0.615}(\Omega_{\rm DM}/0.26)^{0.5}\times0.527\,\mathrm{erfc}[-1.15\,(T_{\rm QCD}/170\,\mathrm{MeV})^{2.15}$ whose inverted dark-matter line sits at $\sin^2 2\theta \sim 1.23\times10^{-8}$ at 3 keV, 3.0×10^{-9} at 7.1 keV, 1.73×10^{-9} at 10 keV, 5.6×10^{-10} at 20 keV and 1.26×10^{-10} at 50 keV, 10 times at 3 keV to 17 times at 50 keV more mixing than the server line at 1.2×10^{-9}, 2.55×10^{-10}, 1.37×10^{-10}, 3.9×10^{-11} and 7.6×10^{-12}, and 3.4–5.5 times more than the brief's formula. At 7.1 keV the verdict depends on normalization: the server line 2.55×10^{-10} survives the XRISM $\lesssim 4\times10^{-10}$ limit while the brief's 7.65×10^{-10} and the published 3.0×10^{-9} are excluded by about 7.5 times, so with the published normalization Dodelson-Widrow at 7.1 keV is excluded by XRISM alone and the server underestimates the mixing needed by about 12 times.

Structure-formation translation uses the standard Dodelson-Widrow to thermal mapping $m_s = 4.43\,\mathrm{keV}\,(m_{\rm th}/\mathrm{keV})^{4/3}$ from Viel et al. astro-ph/0501562, giving $m_{\rm th} = 0.75$ keV at $m_s = 3$ keV, 1.42 keV at 7.1 keV, 1.84 keV at 10 keV, 3.10 keV at 20 keV and 6.16 keV at 50 keV.

Lyman-α $m_{\rm th} > 4\,(6)$ keV then requires $m_s > 28\,(48)$ keV and satellite counts $m_{\rm th} > 8.4$ keV requires $m_s > 76$ keV; at $m_s \gtrsim 28$ keV the Dodelson-Widrow line has $\sin^2 2\theta \sim 3 \times 10^{-10}$ published or about 10^{-11} server, both far above the NuSTAR ceiling $\sim 10^{-12}$ at 6–40 keV.

A BBN robustness check with linearized slopes $dY_p/dN_{\rm eff} = 0.0135$ and $d{\rm D/H}/dN_{\rm eff} = 0.331 \times 10^{-5}$ about the server Standard Model $Y_p = 0.246815$ and D/H$= 2.4433 \times 10^{-5}$ gives 95% upper bounds $\Delta N_{\rm eff} < 0.32$ Y_p-only, 0.58 [0.44] D/H-only and 0.47 [0.37] combined for brief data $Y_p = 0.245 \pm 0.003$ and D/H$= (2.53 \pm 0.03) \times 10^{-5}$, versus 0.46 Y_p-only, 0.66 [0.46] D/H-only and 0.61 [0.43] combined for server data, with the server tool reporting 0.548; D/H alone prefers $\Delta N_{\rm eff} = 0.26$ for brief data.

3.2 Second wave: BBN validity audit and expanded grids

The second wave re-audited server validity and baselines and expanded the quantitative mapping. A contested audit confirms $m_s <\sim 100$ keV relativistic at BBN, BBN fit calibrated $N_{\rm eff} = 2.044$–6.044, $\omega_b h^2 = 0.019$–0.0257, $\tau_n = 866.4$–890.4 s, exclusion $\Delta\chi^2 > 3.84$, and for $m_s \gtrsim 1$ MeV decay injection not modeled. The brief versus server baseline offset is 0.0001 in Y_p (0.025σ) and 0.047×10^{-5} in D/H (1.88σ on server error); Standard Model $Y_p = 0.24682$ is $+0.48\sigma$ above server Y_p and D/H$= 2.4433 \times 10^{-5}$ is -4.15σ below server D/H driving $\chi^2 = 2.77$, while $\Delta\chi^2$ threshold is baseline-independent. Server caps are reiterated: Shi-Fuller $\Delta N_{\rm eff}$ heuristic, L capped at 0.1, no ξ_e, no reheating, ten tabulated X-ray limits, and the 3.5 keV line at $m_s = 7100$ eV and $\sin^2 2\theta = 7 \times 10^{-11}$ flagged debated.

X-ray radiative-decay predictions for $7.1, 10, 20, 40, 50$ keV at $\sin^2 2\theta = 10^{-10}, 10^{-12}, 10^{-14}$ give $E_\gamma = 3.55$–25 keV, $\Gamma = 2.46 \times 10^{-28}$–$4.25 \times 10^{-24}$ s^{-1} at 10^{-10} scaling as $(m_s/{\rm keV})^5$, lifetimes 4.07×10^{27}–2.35×10^{23} s at 10^{-10}, 4.07×10^{29}–2.35×10^{25} s at 10^{-12} and 4.07×10^{31}–2.35×10^{27} s at 10^{-14}, with Dodelson-Widrow $\Omega_s h^2 = 0.047$ at 7.1 keV, 0.087 at 10 keV, 0.304 at 20 keV overclosing, 1.06 at 40 keV overclosing and 1.58 at 50 keV overclosing at 10^{-10}, all fifteen points BBN-allowed with $\Delta N_{\rm eff} = 6.2 \times 10^{-8}$–$0.00295$ and $\Delta\chi^2 < 0$. The Γ formula is verified within 1–2%: 7.1 keV and 7×10^{-11} predicted 1.74×10^{-28} versus tool 1.72×10^{-28} (1.3%), 7.1 keV and 10^{-10} predicted 2.49×10^{-28} versus 2.46×10^{-28} (1.2%), 10 keV and 10^{-10} predicted 1.38×10^{-27} versus 1.36×10^{-27} (1.4%). The 3.5 keV benchmark $m_s = 7.1$ keV, $\sin^2 2\theta = 7 \times 10^{-11}$, $E_\gamma = 3.55$ keV is flagged excluded by built-in NuSTAR M31 and XMM-Newton blank-sky limits with whole-halo fluxes 3.28 ph s^{-1} cm^{-2} Milky Way center, 4.8×10^{-4} M31, 2.5×10^{-5} Perseus and 2.8×10^{-5} Coma and $\tau = 5.82 \times 10^{27}$ s.

A Dodelson-Widrow grid of seven masses $1, 3, 7.1, 10, 20, 50, 100$ keV times ten mixings 10^{-15}–10^{-6} yields 12/70 BBN-excluded and 31/70 overclosing, with 1 keV and 10^{-6} giving $\Delta N_{\rm eff} = 1.0$, $\Omega_s h^2 = 13.84$ and $\Delta\chi^2 = 23.1$ excluded while 7.1 keV and 10^{-10} gives $\Delta N_{\rm eff} = 6.2 \times 10^{-4}$ and $\Omega_s h^2 = 0.047$ allowed. A 6400-point scan 1–100 keV and 10^{-15}–10^{-6} yields 862/6400 BBN-excluded and 2752/6400 overclosing, with BBN 95% upper limit $\sin^2 2\theta_{\rm max}$ falling from 3.5×10^{-7} at 1 keV to 8.9×10^{-9} at 100 keV (1.23×10^{-7} at 3.6 keV, 7.26×10^{-8} at 6.85 keV, 4.29×10^{-8} at 13.8 keV, 2.54×10^{-8} at 26.2 keV, 1.5×10^{-8} at 49.7 keV). At $\sin^2 2\theta = 10^{-10}$ the X-ray line predictions at 10 keV ($E_\gamma = 5$ keV, $\Gamma = 1.36 \times 10^{-27}$ s^{-1}, $\tau = 7.35 \times 10^{26}$ s, Perseus flux 1.4×10^{-4} ph s^{-1} cm^{-2}) and 20 keV ($E_\gamma = 10$ keV, $\Gamma = 4.36 \times 10^{-26}$ s^{-1}, $\tau = 2.30 \times 10^{25}$ s, Perseus flux 2.24×10^{-3}) are excluded by NuSTAR M31 plus XMM-Newton, while 40 keV ($E_\gamma = 20$ keV, $\Gamma = 1.39 \times 10^{-24}$ s^{-1}, $\tau = 7.18 \times 10^{23}$ s, Perseus flux 3.58×10^{-2}) is not excluded by tabulated limits within 30% in mass. Combined BBN plus X-ray viability at $\sin^2 2\theta = 10^{-10}$ is therefore BBN-allowed with $\Delta\chi^2 = -0.008$ to -0.038 and $\Delta N_{\rm eff} = 0.0006$–$0.0029$ for all five masses but X-ray-excluded at 7.1/10/20 keV, overclosing at 40/50 keV; at 10^{-12} all masses are BBN-allowed and not overclosing with $\Omega_s h^2 = 0.00047$–0.0158 and $\tau = 4.07 \times 10^{29}$–$2.35 \times 10^{25}$ s requiring external 2024–2026 limits to assess exclusion; at 10^{-14} all masses are far below dark-matter abundance and BBN limits.

Data and code availability

All outputs of this session are collected in the project folder `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray`. The session produced 29 data files; the supplement lists them. Full documentation of how these results were obtained is in the supplementary material (`supplement.pdf`: implementation details, the tool-call log, the research board with every finding and its evidence, figures not shown here, and the code index) and the verbatim transcript (`transcript.pdf`), generated alongside this report.